

# Disclaimer

- This training resource was developed to be used with the CZ ID platform
- In October, 2026, stewardship of CZ ID transitioned to UCSF and the platform was renamed to SeqtoID
- These training materials are not being actively maintained and may not reflect differences between CZ ID and SeqtoID, subsequent changes to the platform, or current methods and best practices in metagenomic sequencing and analysis



## CZ ID mNGS Nanopore Pipeline

# mNGS Nanopore Pipeline Highlights

- Developed in collaboration with bioinformatics team from Oxford Nanopore Technologies (ONT)
- Pipeline accurately identifies microbes from a range of samples:
  - Mock microbial communities
  - Cell culture
  - Humans (stool, nasal samples)
  - Insects (mosquitos)
  - Environmental (hot springs)

# mNGS Workflow Overview

Sample  
Collection

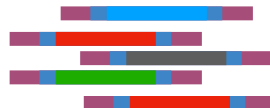
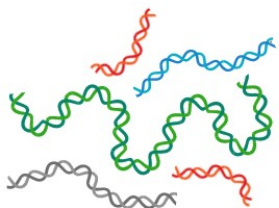
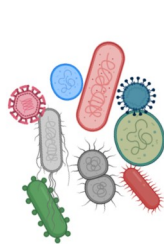
DNA or RNA  
Extraction

Library  
Preparation

Sequencing

Bioinformatic  
Analysis

mNGS



- **Taxonomic Identification**
- Microbiome analysis
- Discovery

# Focus for this workshop

Pathogen identification in stool samples and nasopharyngeal swabs

# Sequence File Formats: FASTA vs FASTQ

>Nucleotide Sequence Name

```
ATTGGCTAATTGGCTAATTGGCTAATTGGCTAATTGG  
CTAATTGGCTAATTGGCTAATTGGCTAATTGGCTAATT  
GGCTAATTGGCTAATTGGCTAATTGGCTAATTGGCT
```

```
@NB501961:14:HM7TLBGX2:1:11102:3233:17234 1:N:0:GATCACCA+GATCACCA  
CATTCGGCTGGGTTTCGTCACCCTGCGGGAAGATGCGGGTCCAGGCGATAGAGGTGCGGAAGCATTGAAGCCCA  
+  
AAAAAEEEEEEEEEE/EEEEEEEEEEEEEEEEEEEE<EEEEEEEE//EAEEEEEEAA/EEE/EEAAEEE<6/
```

# Sequence File Formats

- General
  - FASTA
  - FASTQ (Includes sequence identifier, sequence, and **quality scores** for base calls)
- Nanopore-specific formats
  - FAST5
  - POD5

# Nanopore Specific formats

| Feature            | POD5                                  | Older FAST5                      |
|--------------------|---------------------------------------|----------------------------------|
| File format        | Binary (Apache Arrow-style, columnar) | HDF5-based (hierarchical)        |
| Speed              | High (parallel I/O optimized)         | Slower with random access issues |
| Size               | Smaller (less overhead)               | Larger                           |
| Basecaller support | Dorado, Guppy (via pod5-api)          | Guppy, older tools               |
| Ecosystem          | Actively maintained by ONT            | Being phased out                 |

# mNGS Nanopore Pipeline Workflow

## Upload & Data Pre-processing

Nanopore data  
(FASTQ files)

### Notes:

- Pipeline only accepts **FASTQ** files (FAST5 files must be converted)
- You can upload files through the web app or [command line interface \(CLI\)](#)
- The ONT pipeline does not complete adapter/barcode trimming, so FASTQ files must be trimmed prior to upload
- Only one file per sample:  
Multiple FASTQ files for a given sample will be automatically concatenated if filenames follow a certain format and are associated with the same barcode (see [Automatic Concatenation of Nanopore Files](#) for details)  
  
If filenames do not follow the format recognized by the platform, you need to [concatenate](#) (or combine) them into a single file before upload
- Reads with midread barcodes detected should be filtered prior to upload (see more on the Prepare ONT fastqs for upload to CZ ID slide deck)

# mNGS Nanopore Pipeline Workflow

## Upload & Data Pre-processing

Nanopore data  
(FASTQ files)

Quality, Host &  
Human Filtering  
(fastp &  
minimap2)

Subsampling  
to 1M reads



- Adapter trimming occurred on the basecaller software
- Minimum Q  $\geq 9$
- Length  $\geq 100$
- Complexity > 30%
- Host filtering is splice aware if input is RNA

# mNGS Nanopore Pipeline Workflow

## Upload & Data Pre-processing

## Assembly & Mapping

Nanopore data  
(FASTQ files)

Quality, Host &  
Human Filtering  
(fastp &  
minimap2)

Subsampling  
to 1M reads

Assembly  
(metaFlye)

Contigs

## Contigs



## Notes:

- metaFlye only assembles reads > 1kb  
(in many cases we don't see contigs)

# mNGS Nanopore Pipeline Workflow

## Upload & Data Pre-processing

## Assembly & Mapping

Nanopore data  
(FASTQ files)

Quality, Host &  
Human Filtering  
(fastp &  
minimap2)

Subsampling  
to 1M reads

Assembly  
(metaFlye)

Contigs

Mapping reads to  
contigs  
(minimap2)

Subtraction of  
contig-associated  
reads  
(Samtools)

Unassembled  
reads

## Contigs



## Un-assembled reads



# mNGS Nanopore Pipeline Workflow

## Upload & Data Pre-processing

## Assembly & Mapping

## Species Identification

Nanopore data  
(FASTQ files)

Quality + Host  
Filtering  
(fastp &  
minimap2)

Subsampling  
to 1M reads

Assembly  
(metaFlye)

Subtraction of  
contig-associated  
reads  
(minimap2)

Contigs

Unassembled  
reads

Contig alignment against **NT  
and NR** databases  
(minimap2 & DIAMOND)

Unassembled read alignment  
against **NT** database  
(minimap2)

Contigs  
*aligned to NCBI NT and NR*

Aggregate

Un-assembled reads  
*aligned to NCBI NT only*

Species A

Species B

Species C

# mNGS Nanopore Pipeline Workflow

## Upload & Data Pre-processing

## Assembly & Mapping

## Species Identification

## Report

Nanopore data  
(FASTQ files)

Host + Quality  
Filtering  
(minimap2)

Subsampling  
to 1M reads

Assembly  
(metaFlye)

Subtraction of  
contig-associated  
reads  
(minimap2)

Contigs

Unassembled  
reads

Contig alignment against **NT**  
**and NR** databases  
(minimap2 & DIAMOND)

Unassembled read alignment  
against **NT** database  
(minimap2)

Merge  
results

(Aggregate  
total bases  
per taxon)

Sample Report

# Things to note

- mNGS Nanopore pipeline is limited compared to the Illumina pipeline
  - No QC visualization on CZ ID
  - No background models on CZ ID (more on that later)
  - No consensus genomes on CZ ID
  - No visualization to compare samples (heatmap, trees)

**Heatmap and background model alternative:** [No-code interactive app accessed via github](#)

# CZ ID mNGS Pipeline for ONT review complete!

- Next: QC and Interpretation for ONT

# Resources

- Preprint describing the mNGS Nanopore pipeline and its implementation.
  - [CZ ID: a cloud-based, no-code platform enabling advanced long read metagenomic analysis](#)
- See “how-to” guides in CZ ID’s Help Center:
  - [Upload Nanopore Data](#)
  - [Analyze Nanopore Data](#)
- To learn more about the pipeline, see [Nanopore Pipeline Validation](#) or click [here](#) to view pipeline changes over time.
- “Mid-read” [barcode trimming tool](#)
- [Heatmap visualization and background subtraction tool](#)